TEXORA · LECTURE SHEET

Industrial Air Pollution: Emission and Control Technology

Where industrial emissions come from, how air-quality criteria and emission standards are set, how to measure an emission load, and the control devices — from cyclones to bag filters — that bring it down.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 4 · LEVEL 4 TERM 2

1. Sources and pollutants

In a textile factory the air-pollution inventory is headed by **combustion plant**: boilers raising steam, thermoil heaters, gas/diesel generators and dryers burn fuel and emit **particulate matter** (soot, ash, unburnt carbon), **sulphur dioxide** (from sulphur in fuel), **nitrogen oxides** (from high flame temperatures and fuel nitrogen), **carbon monoxide** (from incomplete combustion) and carbon dioxide. Process sources add **fibre dust** (blowroom, carding, garment trimming), **stenter fume** (vaporised spin finishes, oils and resin components), **singeing and coating vapours**, and **dye-powder dust** in the colour kitchen. Each pollutant behaves differently: coarse dust settles near the source; fine particulates (PM_{10}, and especially $PM_{2.5}$ below 2.5 µm) stay airborne, travel far and penetrate deep into lungs; gases disperse widely and some react in air to form secondary pollutants (Topic 2).

2. Air-quality criteria and how standards are set

Two kinds of standard govern air pollution, and the engineer must not confuse them. **Ambient air-quality standards** limit the concentration in the air people breathe (averaged over stated periods — annual, 24-hour, 8-hour, 1-hour). They are set from **criteria documents** that review the health and environmental evidence: at what concentration, over what exposure time, do effects begin in the most sensitive groups (children, asthmatics, the elderly)? A margin of safety is then applied. The pollutants regulated this way almost everywhere are the **criteria pollutants**: particulate matter (PM_{10}, $PM_{2.5}$), SO_2, NO_2, CO, ozone and lead. **Emission standards**, by contrast, limit what may leave an individual source — a stack concentration (e.g. mg per normal cubic metre) or a mass rate — and are set so that, with all sources complying and normal dispersion, the ambient standards can be met; they also reflect what good control technology can achieve. Compliance is demonstrated by **stack monitoring** (sampling the flue gas under defined conditions) and ambient monitoring around the site. Criteria

setting is thus a chain: *health evidence → ambient criterion → emission limit per source → monitoring and enforcement*.

3. Measuring the emission load

An emission is quantified by concentration × gas flow. Concentrations in flue gas are reported per **normal cubic metre (Nm³)** — gas volume corrected to standard conditions — so that hot, expanded gas cannot "dilute" the reading.

$$\text{Emission rate (kg/h)} = \text{Flow (Nm}^3\text{/h)} \times \text{Concentration (mg/Nm}^3\text{)} \div 10^6$$

WORKED EXAMPLE — PARTICULATE LOAD FROM A BOILER

A boiler's flue-gas flow is 20,000 Nm³/h with a measured particulate concentration of 5,000 mg/Nm³.

Emission rate = 20,000 × 5,000 ÷ 10^6 = **100 kg of particulate per hour**.

Control options applied to the same inlet (removal efficiency applied to both concentration and rate):

- Cyclone at 85 % efficiency → outlet 5,000 × 0.15 = **750 mg/Nm³**, rate **15 kg/h**.
- Electrostatic precipitator at 99 % → outlet **50 mg/Nm³**, rate **1.0 kg/h**.
- Bag filter at 99.6 % → outlet **20 mg/Nm³**, rate **0.4 kg/h**.

Lesson: percent efficiency sounds similar (85 vs 99.6) but the *remaining* load differs 37-fold; judge a device by its outlet concentration, not its headline efficiency.

4. Particulate control technology

DEVICE	PRINCIPLE	SUITED PARTICLE SIZE	TYPICAL EFFICIENCY	STRENGTHS / LIMITS
Settling chamber	Gravity settling in an enlarged, slow duct	Coarse (> ~50 µm)	Low	Simple, no power; only a pre-cleaner
Cyclone	Centrifugal force in a vortex throws particles to the wall	> ~5–10 µm	70–90 %	Cheap, robust, dry recovery; poor on fine dust
Wet scrubber	Gas washed by water sprays/droplets that capture particles (and some gases)	> ~1 µm	85–95 %+	Also absorbs gases, handles hot/wet gas; creates a wastewater stream

Electrostatic precipitator (ESP)	Particles charged in a corona field and collected on plates	Fine, incl. < 1 µm	95–99.5 %	Low pressure drop, huge flows; high capital cost, dust-resistivity sensitive
Bag (fabric) filter	Gas passes through fabric; dust cake filters further	Fine, incl. < 1 µm	99–99.9 %	Highest dry efficiency; bags dislike heat, moisture and sparks

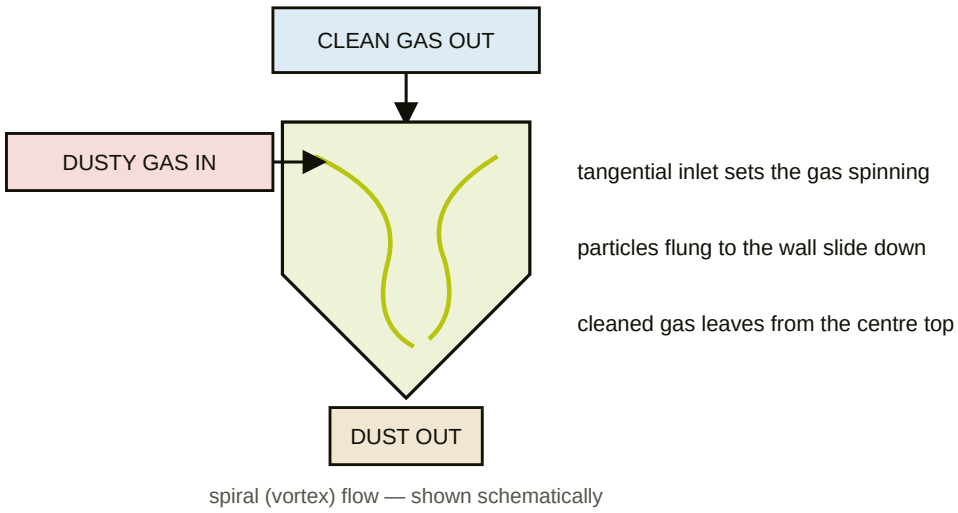


Figure 1 — Cyclone separator. Gas enters tangentially and spins; centrifugal force drives particles to the wall, from where they slide to the dust outlet, while cleaned gas exits through the central top pipe. Efficient on coarse dust, weak on fine particles — often a pre-cleaner before a bag filter or ESP.

Device selection weighs particle size, gas volume and temperature, required outlet concentration, whether the dust is worth recovering dry, and running cost (pressure drop = fan energy). In practice devices are combined: a cyclone ahead of a bag filter removes the coarse load and protects the bags.

5. Gaseous-pollutant control and source measures

Gases are controlled by **absorption** (scrubbing into a liquid — alkali scrubbing for SO_2 and acid gases), **adsorption** (activated carbon for solvent vapours and odours, regenerated or replaced), **combustion/after-burning** (VOC-laden stenter exhaust oxidised at high temperature), and **conversion** (catalytic reduction of NO_x). Cheaper than any of these is **not generating the pollutant**: correct fuel choice (low-sulphur fuel cuts SO_2 at source), burner tuning and excess-air control (cuts CO and NO_x), preventive maintenance, and enclosure with extraction so that fugitive emissions become a controlled, treatable stream. Finally, **dispersion** through an adequate stack height lowers ground-level concentration around the plant — it protects the neighbourhood but does not reduce the total load, so regulators treat it as a supplement to control, never a substitute.

Glossary

Criteria pollutants — the small set of widespread pollutants (PM, SO_2, NO_2, CO, O_3, Pb) for which ambient standards are set from health criteria. **Ambient standard** — limit on concentration in breathed air over a stated averaging time. **Emission standard** — limit on what a source may discharge. **Nm³** — normal cubic metre, gas volume at standard conditions. **PM_{10} / $PM_{2.5}$** — particulate matter below 10 / 2.5 µm aerodynamic diameter. **Cyclone** — vortex separator for coarse dust. **ESP** — electrostatic precipitator. **Bag filter** — fabric filtration unit. **Scrubber** — wet gas-cleaning device using liquid sprays. **Fugitive emission** — leakage not passing through a stack or vent.

Key points

• Combustion plant dominates the textile air inventory; process dust and stenter fume follow.
• Standards form a chain: health criteria → ambient standards → source emission limits → monitoring.
• Emission rate = flow × concentration; in the example, 20,000 Nm³/h at 5,000 mg/Nm³ is 100 kg/h.
• Judge collectors by outlet concentration: cyclone 750, ESP 50, bag filter 20 mg/Nm³ on the same inlet.
• Gases are absorbed, adsorbed, burnt or converted — but fuel choice, tuning and enclosure prevent them cheapest; dispersion dilutes, it does not remove.

Practice

1. A stack discharges 15,000 Nm³/h of gas at 8,000 mg/Nm³ particulate. Compute the inlet load in kg/h. An ESP of 99.2 % efficiency is fitted: find the outlet concentration and the remaining load.
2. Distinguish ambient air-quality standards from emission standards, and outline the chain by which criteria become an enforceable limit on one boiler.
3. Why is a cyclone often installed before a bag filter even though the bag filter alone is more efficient?
4. State the principle, best particle range and one limitation each of a wet scrubber and an ESP.
5. "A taller stack is a pollution-control device." Discuss in four or five sentences.

Answer key

1. Inlet load = 15,000 × 8,000 ÷ 10^6 = **120 kg/h**. Outlet concentration = 8,000 × (1 − 0.992) = 8,000 × 0.008 = **64 mg/Nm³**. Remaining load = 15,000 × 64 ÷ 10^6 = **0.96 kg/h**.
2. Ambient standards limit concentrations in the air people breathe, set from health criteria with a safety margin, over stated averaging times. Emission standards limit an individual source's discharge (concentration or rate). Chain: health evidence is reviewed in criteria documents → ambient

standard fixed → emission limits allocated to source categories so ambient standards can be met with all sources complying, reflecting achievable control technology → limits enforced by stack monitoring.

3. The cyclone removes the coarse fraction (about 70–90 % of the mass) cheaply and robustly, so the bag filter receives a much smaller, finer load: bags last longer, cleaning cycles and pressure drop fall, and spark/coarse-particle damage risk drops. The combination reaches bag-filter outlet quality at lower running cost than bags alone on raw gas.
4. Wet scrubber: gas is contacted with liquid droplets that capture particles (and soluble gases); best above ~1 µm; limitation — it converts an air problem into a wastewater stream needing treatment (also acceptable: corrosion/energy for high efficiency). ESP: particles are electrically charged and collected on plates; excellent on fine particles below 1 µm at large flows; limitation — high capital cost and performance depends on dust resistivity (also acceptable: poor on gaseous pollutants).
5. A taller stack releases the same mass of pollutant but from a greater height, so the plume disperses over a larger volume and ground-level concentrations near the plant fall. It therefore protects local receptors and can help meet ambient standards. It does not reduce total emissions, long-range transport or deposition, and regulators accordingly require source control first, treating stack height as supplementary.

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